

Gautham Chandra

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Education

Bachelor of Science in Aerospace Engineering | University of Oklahoma, Norman

Expected May 2026

Relevant Coursework: Aerodynamics, Aircraft Performance, Flight Mechanics, Flight Dynamics & Controls, Aerospace Structures, CFD

Engineering Experience

Solakair | Aerospace Engineering Intern

Feb 2026 – May 2026

- Accomplished **improved airframe manufacturability and integration robustness** as measured by **fewer fit-up deviations at assembly** by doing a **wing split redesign** in **SolidWorks**, defining datum structure, mating features, and interface controls for repeatable installation.
- Accomplished **faster design iteration** as measured by **shorter CAD release/update cycles** by doing configuration-controlled model revisions, documenting **as-designed vs. as-built** deltas, and supporting repeatable interface updates across the wing assembly.
- Accomplished **faster electrical integration troubleshooting** as measured by **reduced bench bring-up time** by doing harness documentation, verifying pinouts and continuity, and standardizing connectors/terminations for reliable installation and test readiness.

OU Design Build Fly (DBF) | Lead Aerodynamics Engineer

Aug 2023 – Present

- Accomplished **faster aerodynamic design convergence** as measured by **fewer redesign loops** by doing requirements-driven trade studies, **hand calculations**, and screening **400+ airfoils** to down-select a manufacturable wing configuration.
- Accomplished **improved low-speed lift capability** as measured by a **13% increase in section C_l at 0° AoA** by designing a **custom transition airfoil** between **Eppler 398** and **NACA 23012** using interpolation, manual refinement, and XFLR5 validation.
- Accomplished **validated aircraft performance predictions** as measured by **closure of takeoff, landing, climb, and cruise constraints** by generating aerodynamic polars and performance charts in **XFLR5/AVL** using low-fidelity **VLM/panel-method analysis**.
- Accomplished **drag-driver elimination** as measured by a **15% reduction in parasitic drag** by performing **high-fidelity full-aircraft ANSYS CFD** to isolate separation drivers and refine external geometry within competition constraints.
- Accomplished **stronger simulation-to-test correlation** as measured by **agreement between wind-tunnel trends and model predictions** by doing configuration-controlled testing across AoA sweeps, using tuft flow visualization, and updating the aerodynamic model.
- Accomplished a **lighter, stiffer wing build** as measured by a **500 g weight reduction** by doing producibility-focused part breakdown with **3D-printed ribs** and a **carbon-fiber wet-layup spar** while maintaining rigidity.

Undergraduate Research Assistant | AIROU Research Lab, University of Oklahoma

Mar 2025 – Present

- Accomplished **more reliable autonomous UAV operation** as measured by **improved command and telemetry consistency** by integrating **ROS2** perception, navigation, and control nodes with onboard compute.
- Accomplished **faster root-cause identification during flight testing** as measured by **reduced debug cycles** by analyzing telemetry in **Python/MATLAB** to isolate sensor noise, timing offsets, and configuration issues affecting vehicle performance.
- Accomplished **better test readiness and technical communication** as measured by **clearer integration handoff and verification records** by documenting build configuration, wiring/interface details, and test observations for cross-functional team use.

Skills & Achievements

Aerodynamics / Flight Mechanics: Aircraft aerodynamics, flight mechanics, aircraft performance predictions, hand calculations, airfoil analysis, aerodynamic polars, stall analysis, static stability margins, wind-tunnel test data reduction

CFD / Simulation: ANSYS CFD, **STAR-CCM+ (familiar)**, XFLR5, AVL, ANSYS Mechanical (FEA), low- and high-fidelity simulations, VLM/panel methods

Programming: Python, MATLAB, C, Java, HTML/CSS, Firebase

CAD / Design: SolidWorks, NX, CATIA V5, CREO, Autodesk Inventor, GD&T, configuration control, drawing release

Testing / Manufacturing: Wind-tunnel testing, tuft flow visualization, composites (wet layup), 3D printing, additive/subtractive manufacturing, fit-up checks

Robotics / Embedded: ROS2, Pixhawk, SpeedyBee stack, Arduino, Raspberry Pi, UAV mission planning software

Communication: Technical documentation, test reporting, design reviews, cross-functional collaboration

Certifications: FAA Part 107 Remote Pilot Certification

Recognitions: Best Education Project (*Hacklahoma*), Volunteering Presidential Award, Top 25 Internationally (DBF)